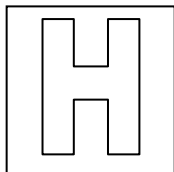


Name:	Index No.:	CT Group: 12
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PIONEER JUNIOR COLLEGE

2013 JC2 Preliminary Examination
HIGHER 1

CHEMISTRY
Paper 2

8872/02

18 September 2013

Candidates to answer Section **A** on the Question Paper

Additional Materials: Data Booklet
 Writing Paper
 Graph Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CT group on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Section B

Answer any **two** questions on separate answer paper.

You are advised to show all working in calculations.
You may use a calculator.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

FOR EXAMINER'S USE			
Section A		Section B	
1	/ 9	6	/ 20
2	/ 7	7	/ 20
3	/ 4	8	/ 20
4	/ 10	Penalty	s.f. / units
5	/ 10	Total	/ 80

Section A (40 marks)

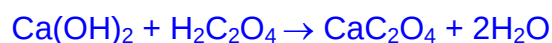
Answer **all** questions in the spaces provided.

- 1** Calcium ethanedioate, CaC_2O_4 , is a white needle-like crystalline solid. It can be prepared by reacting calcium hydroxide, $\text{Ca}(\text{OH})_2$, and ethanedioic acid, $\text{H}_2\text{C}_2\text{O}_4$.

- (a) (i) Name the reaction between calcium hydroxide, $\text{Ca}(\text{OH})_2$, and ethanedioic acid, $\text{H}_2\text{C}_2\text{O}_4$.

Acid-base or neutralisation

- (ii) Write an equation for the reaction.



[2]

- (b) When a pure sample of solid anhydrous CaC_2O_4 was heated strongly at 400°C , a white solid **B** and 0.028 g of carbon monoxide gas were obtained as the **only products**.

- (i) Given that 1 mole of CaC_2O_4 decomposes to give 1 mole of $\text{CO}(\text{g})$, identify solid **B** and write an equation for the reaction.

B is CaCO_3 .



- (ii) Hence, determine the mass of **B** obtained in the reaction.

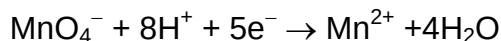
$$\begin{aligned} \text{Amount of B (CaCO}_3\text{)} &= \text{amount CO} \\ &= \frac{0.028}{28.0} \\ &= 0.0010 \text{ mol} \end{aligned}$$

$$\begin{aligned} \text{Mass of B (CaCO}_3\text{) obtained} &= 0.0010 \times 100 \\ &= 0.10 \text{ g} \end{aligned}$$

[4]

- (c) In an experiment, 25.0 cm^3 of $0.050 \text{ mol dm}^{-3} \text{ K}_2\text{C}_2\text{O}_4$ was titrated against $0.022 \text{ mol dm}^{-3}$ acidified KMnO_4 . It was found that 22.70 cm^3 of KMnO_4 was required for complete reaction. During the titration, effervescence was observed and the gas evolved gave a white precipitate in limewater.

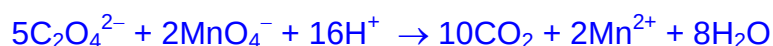
The reaction of manganate(VII) in acidic medium is given below.



- (i) Using the information above, calculate the mole ratio of $\text{C}_2\text{O}_4^{2-}$ to MnO_4^- in the above reaction.

$$\frac{\text{Amount of } \text{K}_2\text{C}_2\text{O}_4}{\text{Amount of } \text{KMnO}_4} = \frac{\frac{25.0}{1000} \times 0.050}{\frac{22.70}{1000} \times 0.022} = 2.5 \text{ or } \frac{5}{2}$$

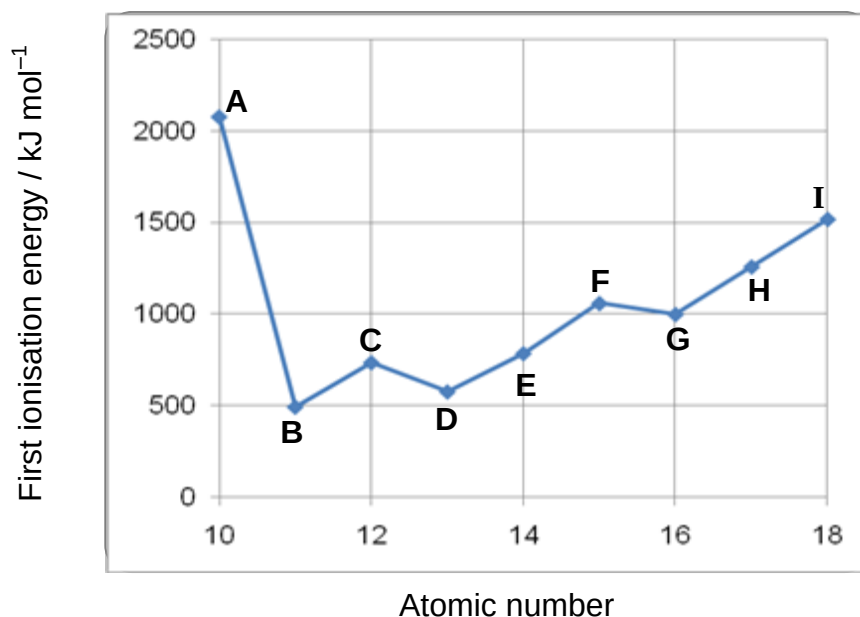
- (ii) Hence, write an equation for the reaction between $\text{C}_2\text{O}_4^{2-}$ and MnO_4^- in acidic medium.



[3]

[Total: 9]

- 2 The diagram below shows a plot of first ionisation energy against atomic number for the elements **A** to **I**. (The letters are not the chemical symbols for the elements concerned.)



- (a) Give the full electronic configuration of the element **D**.



[1]

- (b) Explain briefly why the first ionisation energy of **G** is less than **F**.

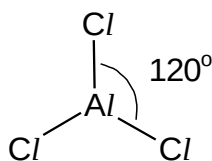
The first ionisation energy of both **F** and **G** involve the removal of an electron from the 3p orbital. However, electron removed from **G** is from a 3p orbital which contains 2 electrons. The paired electrons experience inter-electron repulsion. Thus, less energy is required to remove the 3p electron in **G**.

[2]

- (c) (i) Draw a dot-and-cross diagram to show the electron arrangement in the oxide of **D**.



- (ii) The reaction of **D** with chlorine produces a chloride with a M_r of 133.5. Draw a diagram to illustrate the shape of the chloride of **D**. Indicate clearly the bond angle.



- (iii) In terms of structure and bonding, explain why the oxide of **D** has a higher melting point than the chloride of **D**.

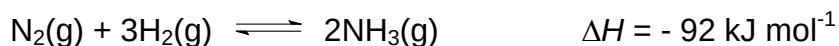
Oxide of **D** has a giant ionic lattice structure whereas the chloride has simple covalent structure.

Oxide of **D** has strong ionic bonds between its oppositely charged ions as compared to the weak intermolecular van der Waals' forces of attraction present in the chloride of **D**. Hence, more energy is required to overcome the stronger bonds in the oxide of **D**.

[4]

[Total: 7]

- 3 Le Chatelier's Principle predicts that the highest equilibrium yield of ammonia in the Haber process,



should occur at high pressure and at low temperature. However, in practice, these conditions are not used.

- (a) What are the typical values of pressure and temperature used industrially? Explain why these conditions are used industrially.

Pressure: 200 – 250 atm

Temperature: 450 – 500 °C

By Le Chatelier's Principle, position of equilibrium of an exothermic reaction shifts to the right at low temperature, giving a high yield. However, the rate of reaction would be too slow at low temperature. Hence in practice, a moderate temperature of 450 °C to 500 °C is adopted.

By Le Chatelier's Principle, increasing pressure will shift equilibrium position to the right, favouring the production of NH₃. The rate of reaction is also faster since concentration and the frequency of effective collisions increases.

However, with increasing pressure, the production cost increases. Therefore, a compromise has to be reached between higher yield and rate at higher cost or lower yield and rate at lower cost. In practice, a high pressure of about 200 - 250 atm is used.

[3]

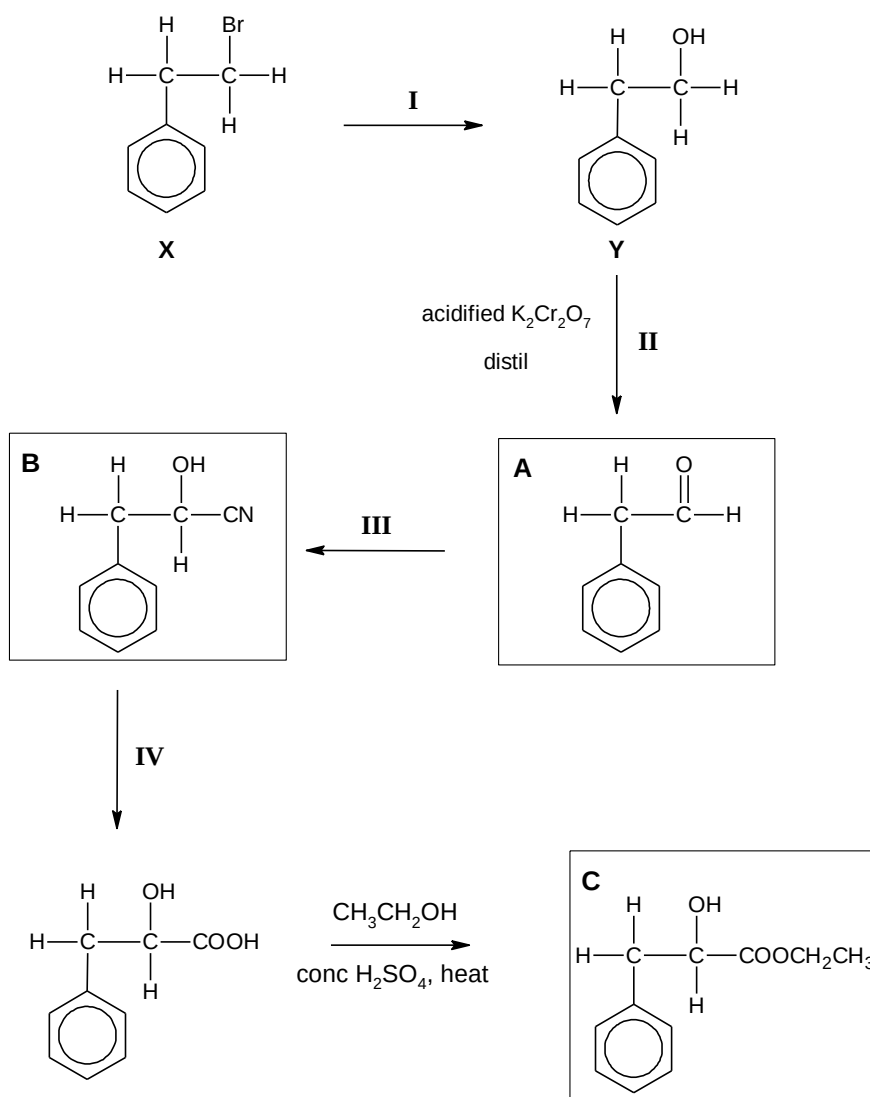
- (b) Why are the gases passed through a tower packed with lumps of iron?

Iron is used as a catalyst to allow equilibrium to be reached at a faster rate.

[1]

[Total: 4]

4 (a) Consider the following reaction scheme.



Draw the structures of the compounds **A**, **B** and **C** in the boxes provided.

[3]

(b) State the type of reaction taking place in steps **I** and **III**.

Step I: (nucleophilic) substitution

Step III: (nucleophilic) addition

[2]

- (c) State the reagents and conditions required for steps I, III and IV.

Step I: NaOH(aq), heat

Step III: HCN, trace NaCN / NaOH, 10 – 20 °C

Step IV: H₂SO₄(aq), heat

[3]

- (d) Suggest a simple chemical test to distinguish compound X from compound Y. You should state clearly the observations for each compound.

Reagent and conditions:

Add NaOH(aq) and heat. Cool the mixture. Acidify with HNO₃(aq). Add AgNO₃(aq).

Observations:

For X, cream precipitate of AgBr observed

For Y, no cream precipitate observed.

OR

Reagent and conditions: Na

Observations:

For X, no effervescence observed.

For Y, effervescence observed. Gas evolved (H₂) which gives a 'pop' sound with lighted splinter.

OR

Reagent and conditions: H₂SO₄(aq), K₂Cr₂O₇(aq), heat

Observations:

For X, orange K₂Cr₂O₇(aq) remains.

For Y, orange K₂Cr₂O₇(aq) turns green.

OR

Reagent and conditions: PCl₅, r.t.p / PCl₃, heat. Place a glass rod dipped into aqueous NH₃ near the mouth of the test-tube.

Observations:

For X, no white fumes observed.

For Y, white fumes observed.

[2]

[Total: 10]

5 Compound **K** has the molecular formula $C_4H_8O_2$.

K decolourises aqueous bromine.

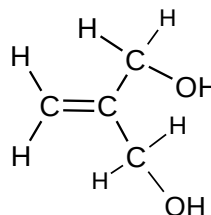
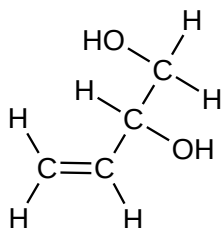
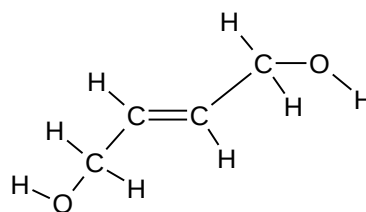
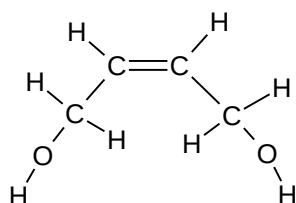
(a) State the functional group in **K** that decolourises aqueous bromine.

alkene or carbon-carbon double bond

[1]

(b) Draw the structural formula of **all** the possible isomers of **K** which have the following features.

- **K** decolourises aqueous bromine.
- **K** reacts with sodium metal but not with aqueous sodium hydroxide.
- No oxygen atom is attached to sp^2 hybridised carbon atom.
- No carbon atom has more than one oxygen atom joined to it.

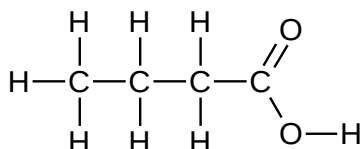


OR

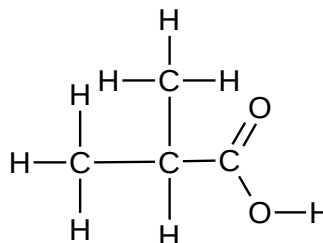
[3]

(c) Compound **K** has another isomer, **J**, which does not decolourise aqueous bromine but react with sodium hydroxide at room temperature.

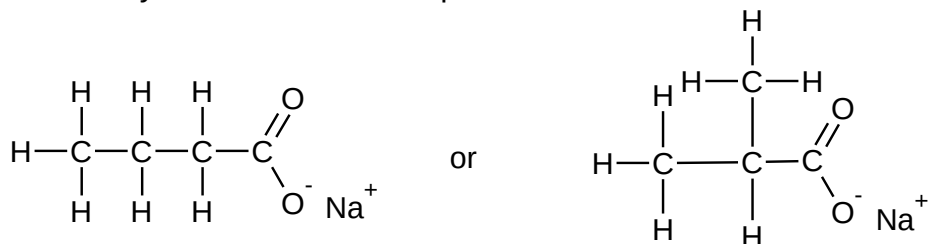
(i) Draw the displayed formula of **J**.



or



- (ii) Give the structure of the organic compound formed when **J** reacts with sodium hydroxide at room temperature.



[2]

- (d) An aqueous solution of an acid HA of concentration $0.100 \text{ mol dm}^{-3}$ has a *pH* of 2.9.

- (i) Define the term in *italics*.

pH is the negative logarithm to base 10 of the hydrogen ion concentration in mol dm^{-3} .

or

$$\text{pH} = -\lg [\text{H}^+] = -\lg [\text{H}_3\text{O}^+]$$

- (ii) Using the information, calculate the K_a of HA.

$$\begin{aligned}
 \text{pH} &= 2.9 = -\lg[\text{H}^+] \\
 [\text{H}^+] &= 1.258 \times 10^{-3} \text{ mol dm}^{-3}
 \end{aligned}$$

	HA(aq)	$\rightleftharpoons$	$\text{H}^+(\text{aq})$	+	$\text{A}^-(\text{aq})$
Initial [] / mol dm^{-3}	0.100		0		0
Change in [] / mol dm^{-3}	-1.258×10^{-3}		$+1.258 \times 10^{-3}$		$+1.258 \times 10^{-3}$
Equilibrium [] / mol dm^{-3}	$0.100 - 1.258 \times 10^{-3}$ ≈ 0.100		1.258×10^{-3}		1.258×10^{-3}

Since HA is a weak acid, the extent of dissociation is very small.

Hence 1.258×10^{-3} is negligible compared to 0.10 mol dm^{-3} .

$$\therefore (0.100 - 1.258 \times 10^{-3}) \approx 0.100$$

$$\begin{aligned}
 K_a \text{ of HA} &= (1.258 \times 10^{-3})^2 / 0.100 \\
 &= 1.58 \times 10^{-5} \text{ mol dm}^{-3}
 \end{aligned}$$

Or

$$\begin{aligned}
 K_a \text{ of HA} &= (1.258 \times 10^{-3})^2 / (0.100 - 1.258 \times 10^{-3}) \\
 &= 1.60 \times 10^{-5} \text{ mol dm}^{-3}
 \end{aligned}$$

[4]

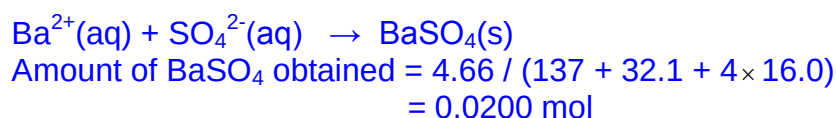
[Total: 10]

Section B (40 marks)

Answer **two** questions from this section on separate answer paper.

- 6 (a) 5.34 g of a salt, M_2SO_4 (where **M** is a metal), was dissolved in water. The sulfate ion was precipitated by adding excess barium chloride solution. 4.66 g of barium sulfate precipitate was obtained.

- (i) Calculate the number of moles of sulfate ions precipitated.



- (ii) Determine the relative formula mass of M_2SO_4 . Hence, calculate the relative atomic mass of metal **M** and state its identity.



$$M_r \text{ of } M_2SO_4 = 5.34 / 0.0200 = \underline{267}$$

$$2(A_r \text{ of } M) + 32.1 + 4(16.0) = 267$$

$$A_r \text{ of } M = \underline{85.5}$$

M is Rubidium (Rb).

[5]

- (b) State and explain whether metal **M** or calcium would have a higher melting point. [3]

Calcium would have a higher melting point than **M** due to smaller cationic radius, hence higher charge density of cation and the stronger attraction for the delocalised electrons, hence the stronger the metallic bond. There are also more valence electrons contributed for metallic bonding in **M**, hence the stronger the metallic bond.

- (c) State another two physical properties of metal **M**. [2]

- Good conductor of electricity due to presence of mobile sea of delocalised electrons.
- High thermal conductivity. When heat is supplied at one end of the metal, kinetic energy of the atoms and electrons is increased. These particles collide, transfer and conduct *heat energy* to surrounding atoms towards the cooler region of the metal.
- Malleable and ductile. If stress is applied to metals, layers slide over each other without breaking the metallic bond.

- (d) Do you expect the chloride of **M** to dissolve in water? Briefly explain your answer. [1]

Chloride of **M** is expected to be soluble in water as the ions can form favourable ion-dipole interactions with water molecules.

- (e) (i) When metal **M** reacts with water, effervescence is observed and an alkaline solution is obtained. Write an equation of the reaction.



- (ii) Explain what is meant by the term *standard enthalpy change of neutralisation*.

Standard enthalpy change of neutralisation (ΔH_n^θ) is the energy evolved when one mole of water is formed from the neutralisation between acid and alkali under standard conditions.

- (iii) 40 cm³ of 3.0 mol dm⁻³ hydrochloric acid was added to 60 cm³ of 1.4 mol dm⁻³ of the metal **M** hydroxide in a polystyrene cup. The maximum temperature rise recorded was 11.6 °C.

Given that the specific heat capacity of the solution = 4.2 J g⁻¹ K⁻¹ and the process is 100% efficient, calculate the enthalpy change of neutralisation for the reaction.

$$\text{Amount of HCl} = 40/1000 \times 3 = 0.1200 \text{ mol}$$

$$\text{Amount of MOH (base)} = 60/1000 \times 1.4 = 0.08400 \text{ mol}$$

MOH is the limiting reagent.

$$\text{Amount of H}_2\text{O} = 0.08400 \text{ mol}$$

$$\text{Heat evolved} = mc\Delta T = 100 \times 4.2 \times 11.6 = 4872 \text{ J}$$

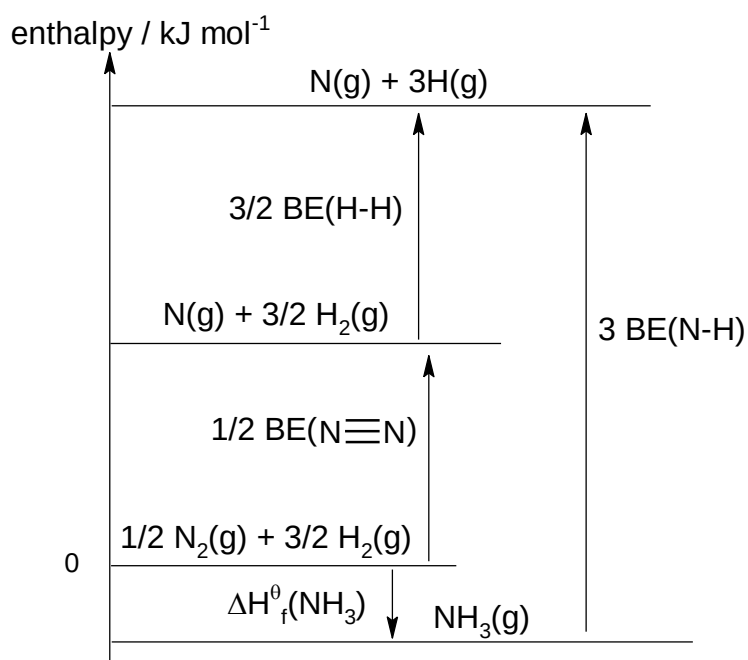
$$\Delta H_n^\theta = -4872 / 0.084 = -58000 \text{ J mol}^{-1} = \underline{-58.0 \text{ kJ mol}^{-1}}$$

- (iv) The enthalpy change of neutralisation of ammonia and hydrochloric acid is – 55.2 kJ mol⁻¹.

Account for the difference in the values of the enthalpy change of neutralisation.

The enthalpy change of neutralisation involving HCl and NH₃ is less exothermic. NH₃ is a weak base but RbOH is a strong base. Some of the energy evolved from the neutralisation process is used to further dissociate the weak base completely / to cause ionisation of the undissociated weak base.

- (f) The enthalpy change of formation of ammonia gas is -46 kJ mol^{-1} . Using the following energy level diagram and other data from the *Data Booklet*, calculate the bond energy of N–H bond in ammonia.



By Hess' Law,

$$3\text{BE}(\text{N-H}) = \frac{3}{2}\text{BE}(\text{H-H}) + \frac{1}{2}\text{BE}(\text{N}\equiv\text{N}) - \Delta H_f^\circ(\text{NH}_3)$$

$$= \frac{3}{2}(436) + \frac{1}{2}(994) - (-46) = 1197$$

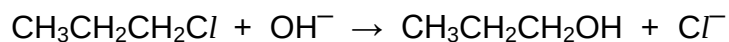
$$\text{BE}(\text{N-H}) = 1197 / 3 = \underline{+399 \text{ kJ mol}^{-1}}$$

[2]

[Total: 20]

- 7 (a) Halogenoalkanes are useful intermediates in organic synthesis. They can undergo hydrolysis with a base to produce alcohols.

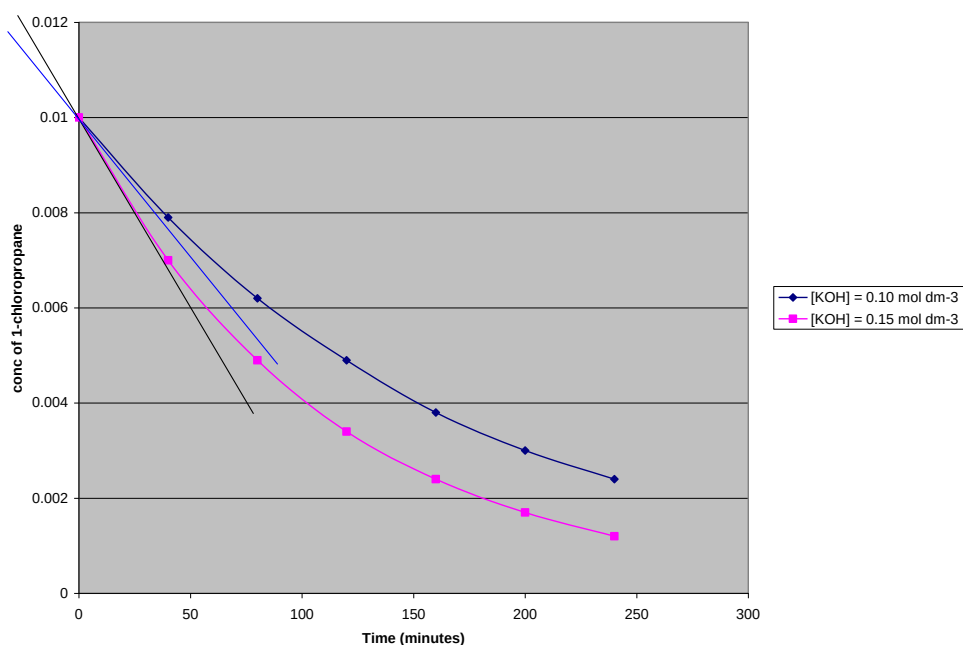
The kinetics of the hydrolysis of 1-chloropropane with aqueous potassium hydroxide may be investigated by determining the concentration of 1-chloropropane remaining at different time intervals. The reaction is represented by the equation below.



The following results were obtained from two experiments on such a hydrolysis. The reaction was followed twice with different concentrations of potassium hydroxide.

Time / minutes	Experiment 1 [KOH] = 0.10 mol dm ⁻³	Experiment 2 [KOH] = 0.15 mol dm ⁻³
	[1-chloropropane] / mol dm ⁻³	[1-chloropropane] / mol dm ⁻³
0	0.0100	0.0100
40	0.0079	0.0070
80	0.0062	0.0049
120	0.0049	0.0034
160	0.0038	0.0024
200	0.0030	0.0017
240	0.0024	0.0012

- (i) Using the same axes, plot graphs of [1-chloropropane] against time for the two experiments, labelling each graph clearly.



- (ii) Use your graph to deduce the order of reaction with respect to 1-chloropropane. Show your working clearly.

From the graph, determine

The half-life for 2nd experiment is 78 min.

OR

The half-life for 1st experiment is 118 min.

As the half-life is a constant, the order of reaction with respect to 1-chloropropane is one.

- (iii) Use your graph to determine the initial rate of each reaction. Hence, use it to deduce the order of reaction with respect to potassium hydroxide.

Draw tangent to the two curves at $t = 0$ min.

Initial rate for $[\text{KOH}] = 0.10 \text{ mol dm}^{-3}$ is $5.25 \times 10^{-5} \text{ mol dm}^{-3} \text{ min}^{-1}$

Initial rate when $[\text{KOH}]$ is 0.15 mol dm^{-3} is $8.00 \times 10^{-5} \text{ mol dm}^{-3} \text{ min}^{-1}$

$[\text{1-chloropropane}]$ kept constant at $t = 0$ min.

When $[\text{KOH}]$ is increased by 1.5 times, the rate increases by 1.5 times.

Thus, the order of reaction with respect to KOH is one.

- (iv) Construct a rate equation for the reaction and use it to calculate the rate constant, giving its units.

$$\text{Rate} = k[\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}][\text{KOH}]$$

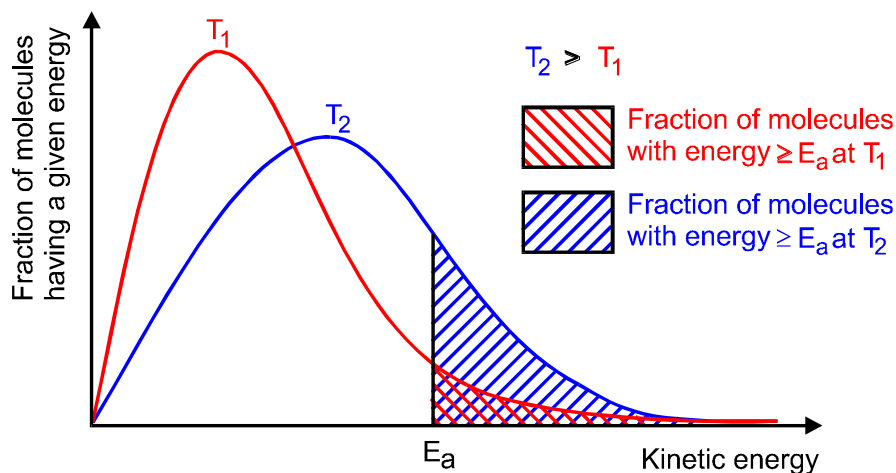
Substitute initial rate = $5.25 \times 10^{-5} \text{ mol dm}^{-3} \text{ min}^{-1}$, $[\text{OH}^-] = 0.1 \text{ mol dm}^{-3}$ and $[\text{1-chloropropane}] = 0.01 \text{ mol dm}^{-3}$ into the rate equation,

$$k = 0.0525 \text{ mol}^{-1} \text{ dm}^3 \text{ min}^{-1}$$

- (v) Explain, with an aid of the Maxwell-Boltzmann distribution curve, the effect of increasing temperature on the rate of the reaction.

[10]

Maxwell-Boltzmann distribution curve

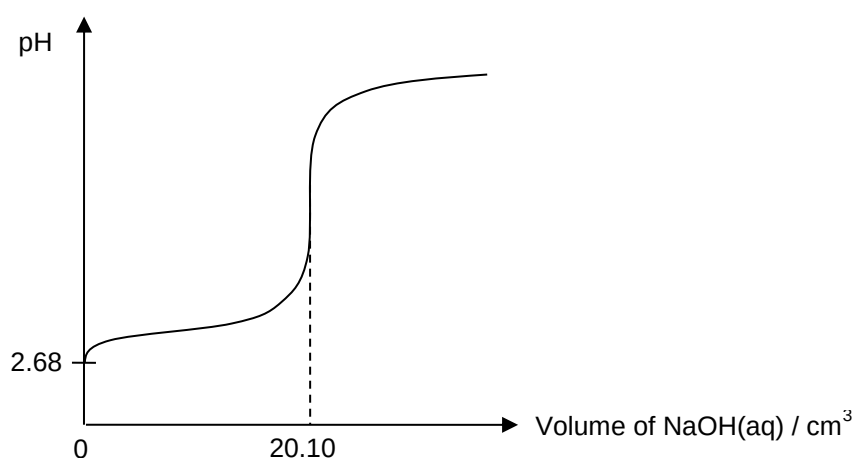


When temperature is increased, there is

- an increase in the average kinetic energy of the particles, resulting in,
- an increase in the number of reacting particles with energy $\geq E_a$,
- an increase in the frequency of collisions between reacting particles,
- resulting in an increase in effective collisions per unit time,
- an increase in rate of reaction.

Note: At higher temperature T_2 , the curve shifts right with a lower peak since average kinetic energy of particles increase. The total number of particles (total area under the curve) remains the same.

- (b) Acrylic acid, $\text{CH}_2=\text{CHCOOH}$, is a monobasic acid. 25.0 cm^3 of the acid required 20.10 cm^3 of 0.10 mol dm^{-3} NaOH(aq) for complete reaction, as shown by the following titration curve.



- (i) Use the titration curve to calculate the concentration of acrylic acid used.

$$\text{Amount of NaOH} = 0.10 \times 0.02010 = 2.01 \times 10^{-3} \text{ mol}$$

$$[\text{acrylic acid}] = 2.01 \times 10^{-3} / 0.025 = 0.0804 \text{ mol dm}^{-3}$$

- (ii) Using the pH value of the acrylic acid shown in the graph, calculate the concentration of the hydrogen ions in the solution of acrylic acid. Hence, deduce whether acrylic acid is a weak or strong acid.

$$[\text{H}^+] = 10^{-2.68} = 2.09 \times 10^{-3} \text{ mol dm}^{-3}$$

$$[\text{H}^+] \ll [\text{acid}], \text{ acrylic acid is weak.}$$

- (iii) Suggest, with reason, a suitable indicator for the titration of acrylic acid with sodium hydroxide.

Phenolphthalein

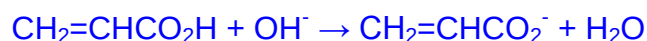
There is sharp / distinct colour change at end point from colourless to pale pink.

OR

Working range of the indicator pH 8 to 10 coincides with the equivalence point of titration.

- (iv) Before the end-point is reached, the pH of the solution changes gradually as shown in the graph. This is due to the presence of a mixture of acrylic acid ($\text{CH}_2=\text{CHCO}_2\text{H}$) and its salt ($\text{CH}_2=\text{CHCO}_2\text{Na}$) in the solution.

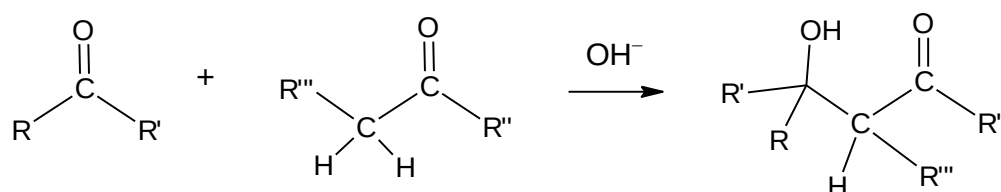
By means of an ionic equation, show how this mixture of acrylic acid and its salt removes the small amount of sodium hydroxide added as the titration proceeds.



[6]

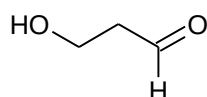
- (c) The aldol reaction is a useful method for making new carbon–carbon bonds in organic chemistry when carbonyl compounds react under suitable conditions.

In alkaline medium, two carbonyl compounds undergo aldol reaction as shown below.

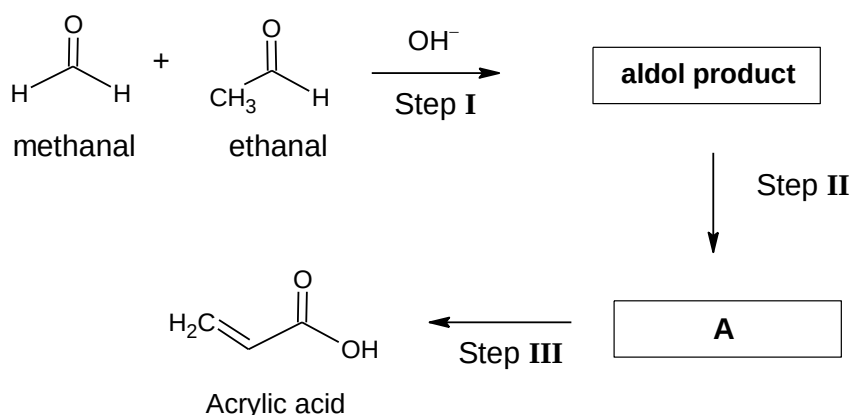


[R groups can be alkyl, aryl or H]

- (i) Draw the structure of the aldol product formed between methanal and ethanal.



- (ii) Acrylic acid can be synthesised from the following reaction scheme.

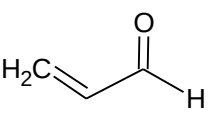


17

Give the structure of compound **A** formed and state the reagents and conditions for Step **II** and **III**.

Step **II**: excess concentrated H_2SO_4 , 170°C

Step **III**: $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, heat

Structure of **A**: 

[4]

[Total: 20]

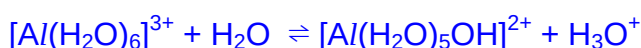
- 8 (a) The chlorides of elements sodium to phosphorus all dissolve or react with water.

Describe the reaction of chlorides of aluminium and phosphorus with water. Suggest the pH values of the resulting solutions and explain your answer with equations. [5]

AlCl_3 dissolves in water and undergo hydrolysis



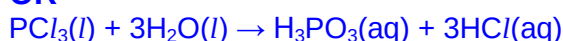
Al^{3+} has high charge density. It can polarise and weaken the O-H bond in water molecules causing them to break, releasing H^+ . Thus, the solution has a pH of 2 – 3.



PCl_5 undergoes hydrolysis to form an acidic solution of H_3PO_4 and HCl . Thus, the solution has a pH of 1 – 2.



OR



- (b) (i) Explain what is meant by *first ionisation energy* of an element, X.

First ionisation energy (1^{st} I.E.) of an element is defined as the amount of energy required, measured in kJ mol^{-1} , to remove one mole of electrons from one mole of gaseous X atoms to form one mole of gaseous X^+ ions.



- (ii) State and explain the difference in first ionisation energy between

I aluminium and phosphorus

- Nuclear charge increases as the number of protons increases.
- Electrons are added to the same quantum shell and shielding effect remains approximately constant as the number of filled inner shells remain constant.
- Overall effect is the increase of effective nuclear charge ($Z_{\text{eff}} = Z_{\text{actual}} - \text{shielding effect}$)
- Increase in effective nuclear charge results in an increased attraction between the nucleus and the outermost electron.
- More energy is required to remove the outermost electron.
- First IE of phosphorus is higher than aluminium

II magnesium and aluminium

Electronic configuration of

Mg : $1s^2 2s^2 2p^6 3s^2$

Al : $1s^2 2s^2 2p^6 3s^2 3p^1$

- The outermost electron in Mg is located in the s orbital while the outermost electron in Al is located in the p orbital.
- Within the same principal quantum shell, the p orbitals have a higher energy compared to the s orbital (p electron is held less strongly).
- More energy is required to remove the 3s electrons of Mg than 3p electron of Al.
- The first ionisation energy of Mg is higher than the first ionisation energy of Al.

[5]

- (c) Compound **P** is found to contain 75.3% carbon, 10.6% hydrogen and 14.1% oxygen by mass. The relative molecular mass of **P** is 112.

When **P** is oxidised with acidified potassium manganate(VII), a compound **Q**, $C_4H_6O_3$, is produced together with propanoic acid. Treatment of **P** with phosphorus pentachloride produces no white fumes.

Q produces a pale yellow precipitate when warmed with alkaline solution of iodine, accompanied by a solution which, when treated with dilute hydrochloric acid, gave compound **R**.

R may also be obtained when $NC-CH_2-CN$ is heated with a dilute acid. **R** reacts with excess pentachloride to produce a compound **S**, together with white fumes evolved.

Deduce the structure of compounds **P**, **Q**, **R** and **S** and explain the chemistry of the reactions described above. Write equations where appropriate. [10]

	C	H	O
% by mass	75.3	10.6	14.1
amt	$75.3/12.0 = 6.275$	10.6	$14.1/16 = 0.881$
mole ratio	7	12	1

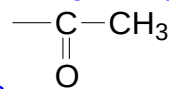
M_r of $(C_7H_{12}O)_n = 112 \Rightarrow n = 1$

P has a molecular formula: $C_7H_{12}O$.

P undergoes oxidation with acidified $KMnO_4$ to produce **Q** and propanoic acid. **P** is likely to contain $C=C$ since 2 products are formed. (not likely to be ester as there is only one oxygen).

P does not undergo (nucleophilic) substitution with PCl_5 . **P** does not contain alcohol functional group.

Q undergoes oxidation with alkaline aqueous iodine. Since **Q** is a product of



oxidation, it cannot contain alcohol group. Thus, it contains_____.

Upon acidification with HCl(aq) , **R** is obtained. Solution contains a carboxylate salt and **R** is carboxylic acid.

$\text{NC-CH}_2\text{-CN}$ undergoes acid hydrolysis with dilute acid to form **R**.

R undergoes (nucleophilic) substitution with PCl_5 to give **S**, which is an acid chloride.

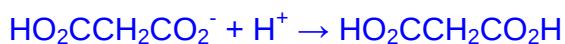
R is $\text{HO}_2\text{CCH}_2\text{CO}_2\text{H}$

Q is $\text{HO}_2\text{CCH}_2\text{COCH}_3$

P is $\text{CH}_3\text{COCH}_2\text{CH=CHCH}_2\text{CH}_3$

S is $\text{ClOCCH}_2\text{COCl}$

Equations:



[Total: 20]

End of Paper